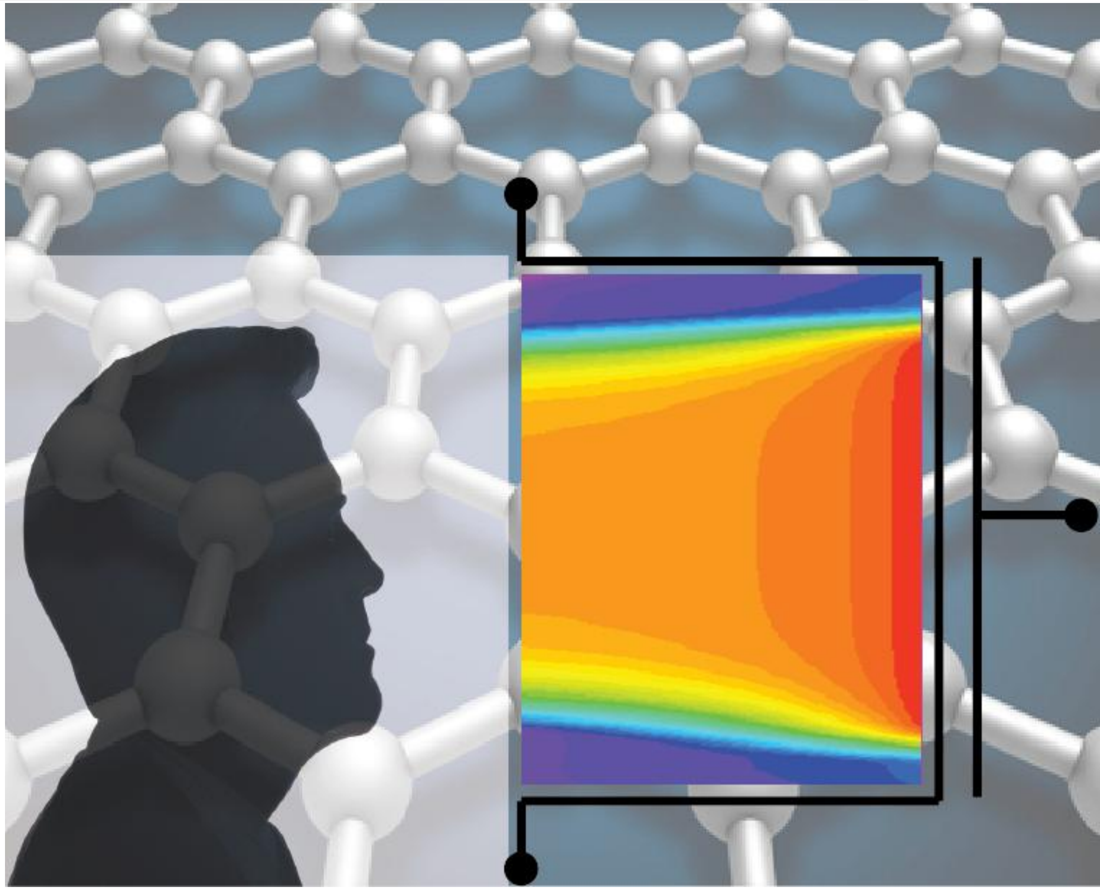
ADVANCED SOLID-STATE DEVICES

Week-8, Lecture-1



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3rd March, 2025

MOSFET

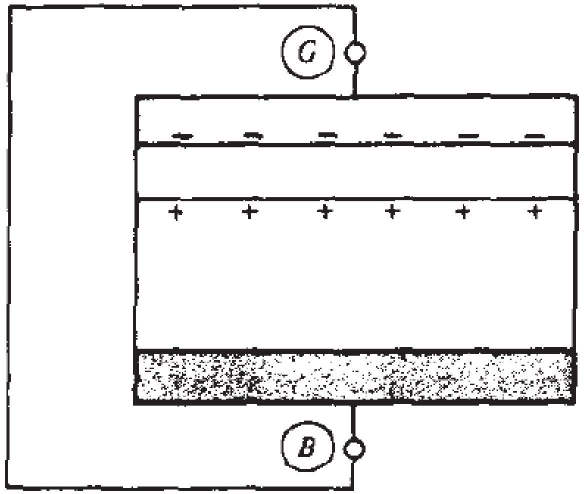
Flat-band Voltage

MOS Capacitor: Flat-band voltage (1)

Consider a MOS structure connected by a external metal wire:

- In going from G through external metal wire and B, we can encounter many contact potential (assuming external metal wire and gate material are different)
- Sum of all contact potentials from G, through external wire, to bulk = $\phi_{gate-material} - \phi_{bulk\ material}$

Existence of nonzero potential between gate and bulk (across oxide) causes net charge to appear on both sides of the oxide



MOS Capacitor: Flat-band voltage (2)

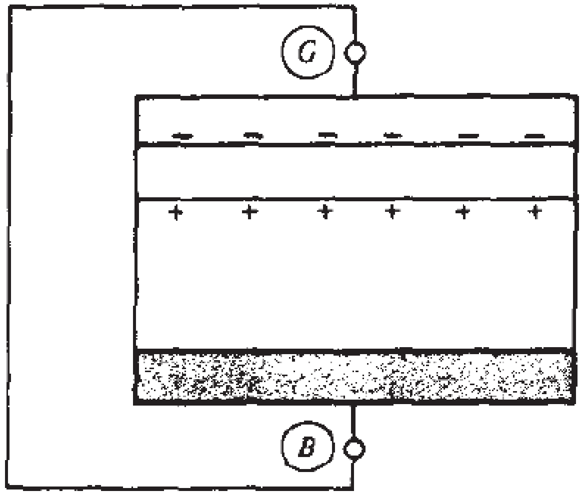
Problem: Consider the MOS structure of the previous slide. Can an external voltage be applied in such a way that the net charges disappear?

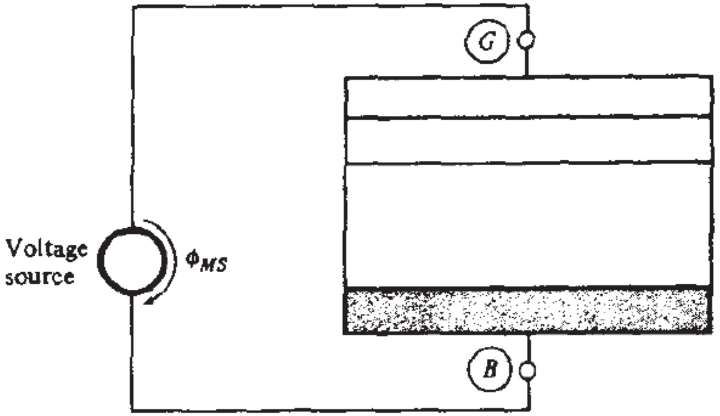
The voltage source should precisely cancel the sum of contact potentials:

$$\phi_{ms} = \phi_{bulk\ material} - \phi_{gate-material}$$

- The condition of disappearance of charge in the semiconductor is known as flat band condition (bands will be flat in the semiconductor)
- The corresponding voltage is known as flat band voltage V_{FB}

- In ideal MOS: $V_{FB} = \phi_{ms}$
- In real devices, in addition to contact potential, there are other factors that result in charge accumulation in semiconductor





MOSFET

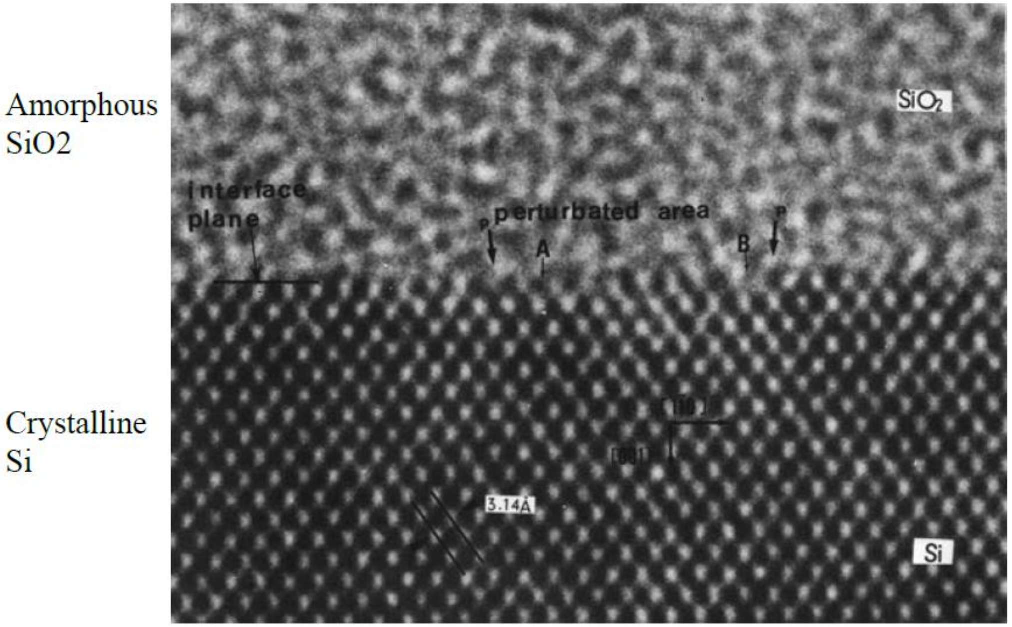
Parasitic Charges

MOS Capacitor: Parasitic Charges (1)

- Parasitic charges exist within the oxide and the oxide-semiconductor interface
- Oxide-silicon interface in real devices are not electrically neutral

- Earlier, the nature and origin of parasitic charges were not well understood
- Inhibited obtaining high-quality MOSFET

- Now, these are well understood and controlled in mature processes
- **Four kind of charges** (nomenclature in 1978 Deal)

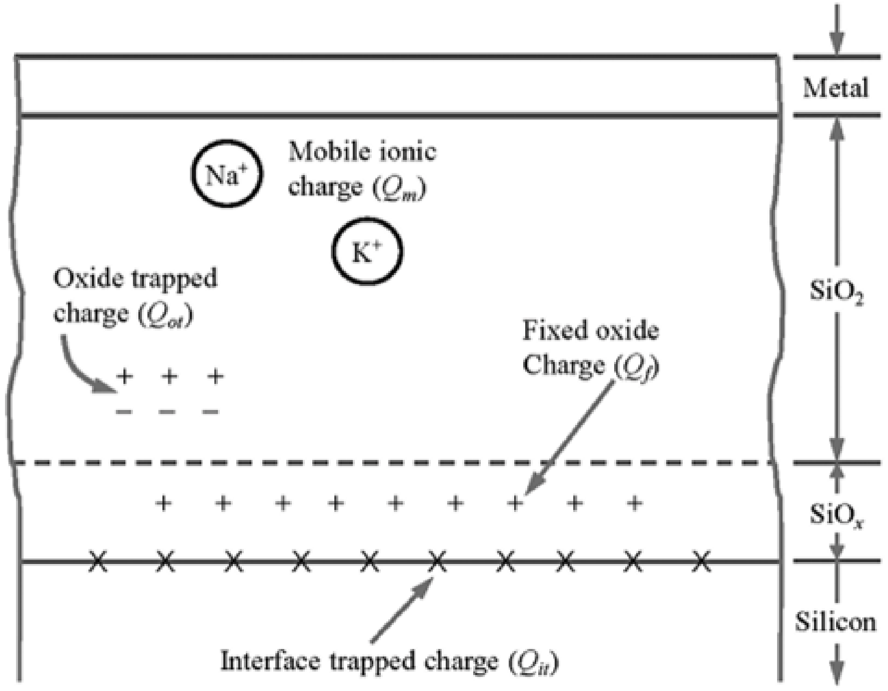


Source: ECE 59500-006 Microfabrication Fundamentals, nanohub.org

MOS Capacitor: Parasitic Charges (2)

1. Mobile Ionic charge: (Q_m)
   - Created by contamination of sodium/potassium ions during fabrication
   - Can respond to electric field and move

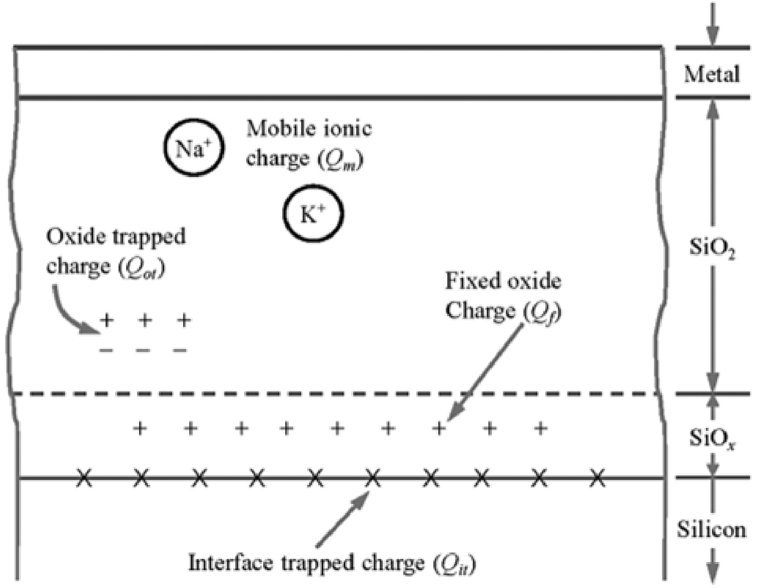
2. Oxide trapped charge: (Q_{ot})
   - Can be distributed inside oxide
   - Acquired due to radiation, injection from substrate



MOS Capacitor: Parasitic Charges (3)

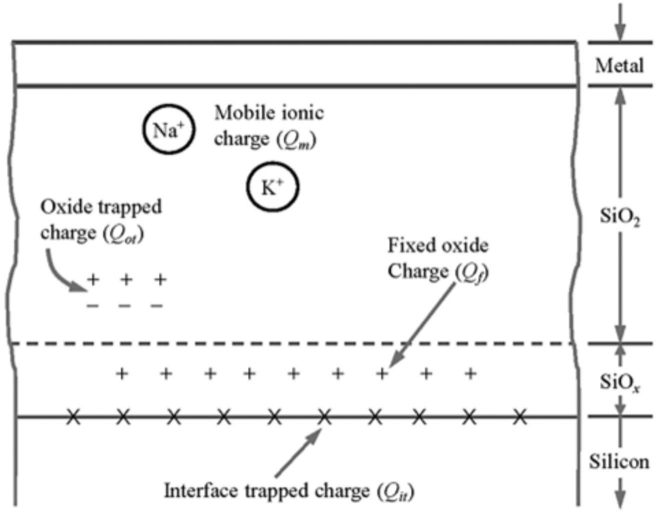
3. Oxide fixed charge: (Q_f)
   - Lie very close to the surface
   - Created due to the mechanism of oxide formation
   - Independent of oxide thickness, doping type, and doping concentration

4. Interface trapped charge: (Q_{it})
   - Exists at the interface due to surface states
   - Occupancy dependent on the energy of the ***surface state*** relative to the Fermi level
   - As surface potential changes, energy level of surface state moves with it
   - Occupancy of surface state change with surface potential
   - Interface trap charges changes with surface potential



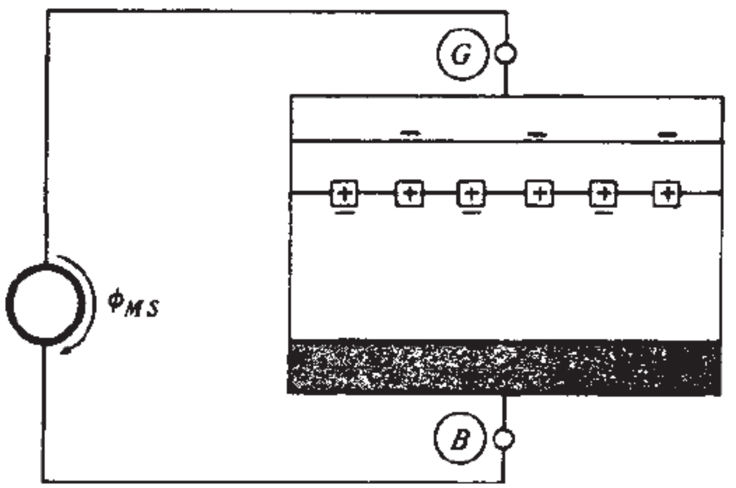
MOS Capacitor: Parasitic Charges (4)

- Effect of arbitrary oxide charge distribution can be accounted for by an equivalent oxide charge located at oxide-silicon interface
- We assume fixed oxide charge density Q'_{ss} at oxide-silicon interface
- Typically $Q'_{ss} < 10^{11}$ electronic charge per cm^2 (it is positive for both p-type and n-type silicon interfaces)

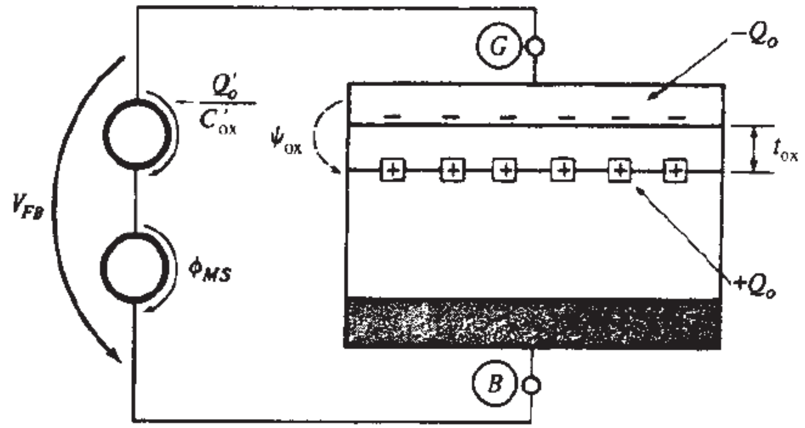


MOS Capacitor: Parasitic Charges and Flat-band voltage

- Consider the MOS structure shown. We have connected a battery with voltage ϕ_{ms} so that the effect of contact potential is nullified.
- Now all the charge that appears at the interface is due to the parasitic charge $+Q_{ss}$
- Charge neutrality requires that a total $-Q_{ss}$ appears



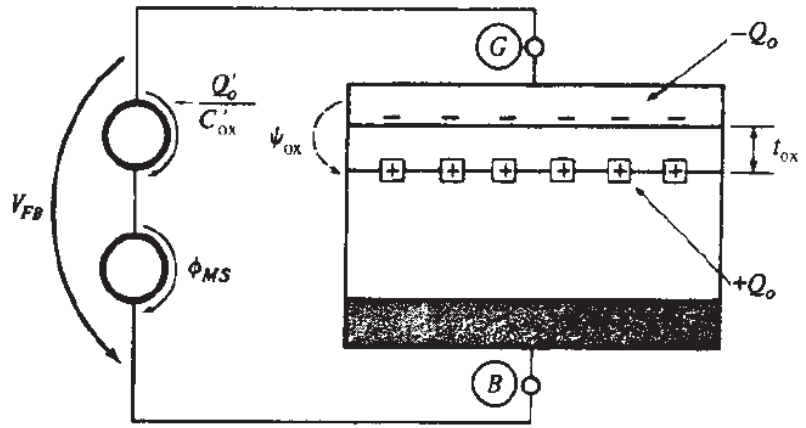
- If we want to eliminate all charge from semiconductor, we need to provide entire $-Q_{ss}$ charge from the battery to the **metal**
- Hence, we need to apply a voltage source of $\psi_{ox} = -\frac{Q_{ss}}{C_{ox}}$
- If we take quantities as per unit area: $\psi_{ox} = -\frac{Q'_{ss}}{C'_{ox}}$ [where $C'_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{\epsilon_r \epsilon_o}{t_{ox}}$]



MOS Capacitor: Total Flat-band Voltage

- To obtain flat band in real devices we need to apply a voltage of:

$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C'_{ox}}$$

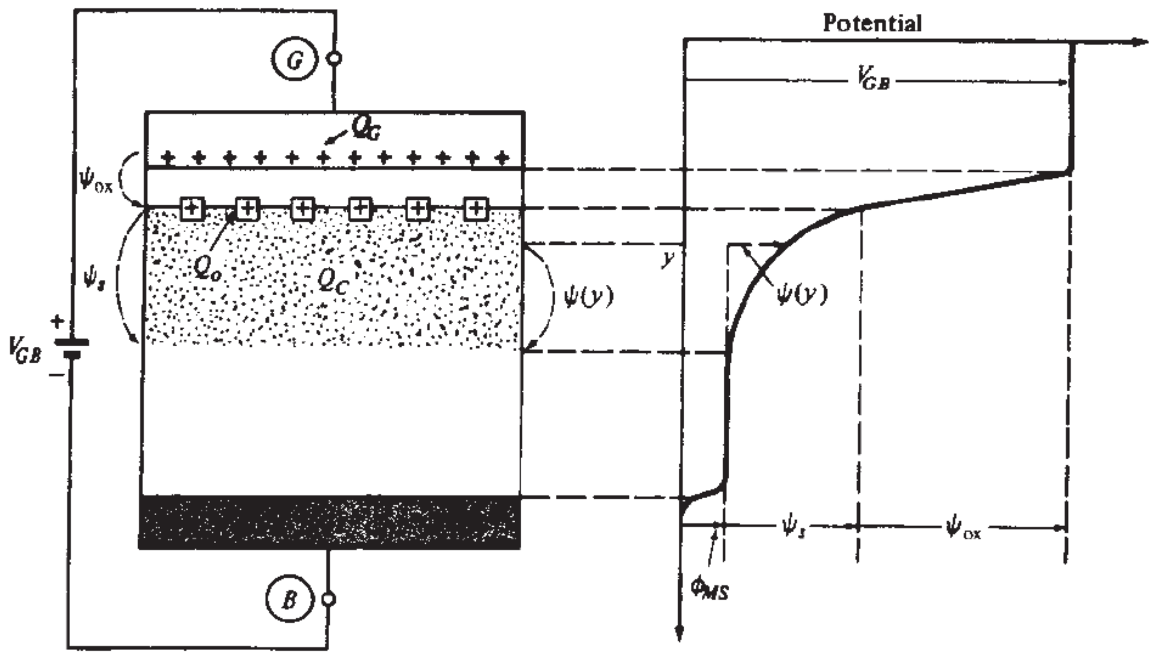


MOSFET

Electrostatics and Threshold Voltage

MOS Capacitor: Potential Balance

- When we apply voltage different than V_{FB}, in general, charge will appear in the semiconductor (mostly close to the semiconductor/oxide interface)



- Under this condition potential and charge balance must be fulfilled

Potential balance:

$$V_{GB} = \phi_{ms} + \psi_s + \psi_{ox}$$

V_{GB}: applied external voltage

ϕ_{ms}: sum of all contact potentials

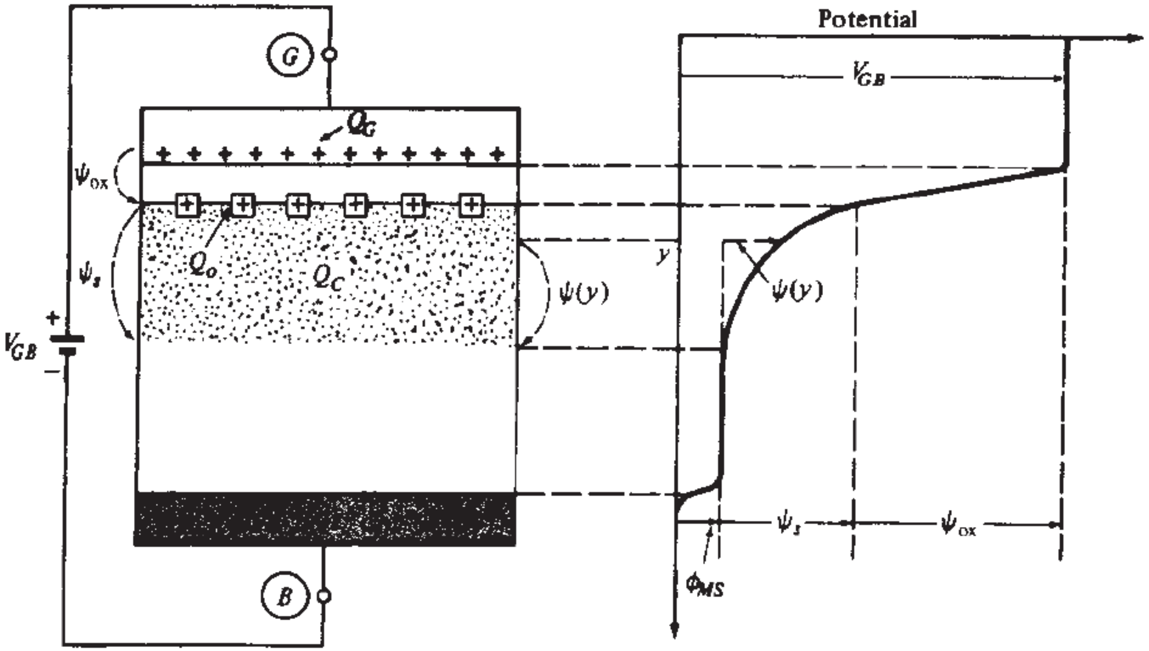
ψ_s: surface potential

ψ_{ox}: potential drop across oxide

Impact of voltage change:

$$\Delta V_{GB} = \Delta\psi_s + \Delta\psi_{ox}$$

MOS Capacitor: Charge Balance



Charge balance:

$$Q_G + Q_{SS} + Q_S = 0$$

Q_G: charge on the gate

Q_{SS}: Effective interface charge (parasitic charge)

Q_S: charge in semiconductor

In terms of per unit area:

$$Q'_G + Q'_{SS} + Q'_S = 0$$

If there is a change in gate charge, it will be balanced by semiconductor charge (and vice versa): $\Delta Q'_G + \Delta Q'_S = 0$

Caution: In books, different notations are used for these potential and charges (physics remains the same)

MOS Capacitor: Threshold Voltage (1)

Threshold Inversion Definition: Gate voltage required such that surface potential $\psi_s = 2\phi_{Fp}$ for p-type semiconductor and $\psi_s = 2\phi_{Fn}$ for n-type semiconductor.

Problem: To compute threshold voltage in terms of device parameters.

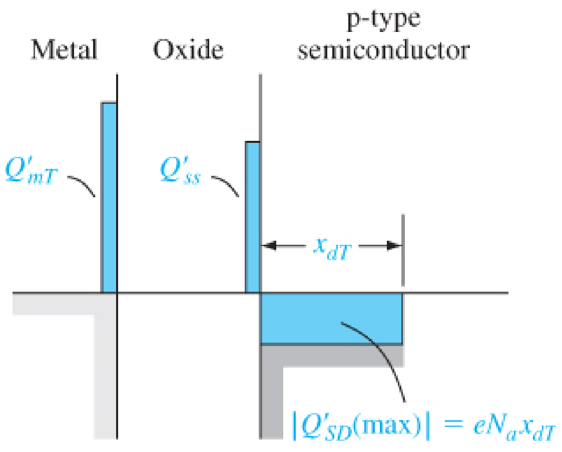


Figure 10.19 | Charge distribution in a MOS capacitor with a p-type substrate at the threshold inversion point.

Charge balance: $Q'_G + Q'_{ss} + Q'_s = 0$

Q'_G: charge on the gate (In Fig. Q'_{mT})

Q'_{ss}: effective interface charge (parasitic charge)

Q'_s : charge in semiconductor. It is mostly due to depletion layer (In Fig. Q'_{SD}). At strong inversion depletion layer reaches maximum $x_{d,max}$ (in Fig. x_{dT}), and total charge is negative charge and equals to $-qN_a x_{d,max}$.

- So: $Q'_G + Q'_{ss} = qN_a x_{d,max} \Rightarrow Q'_G = qN_a x_{d,max} - Q'_{ss}$
- Voltage across oxide: $\psi_{ox} = \frac{Q'_G}{C'_{ox}} = \frac{qN_a x_{d,max} - Q'_{ss}}{C'_{ox}}$

MOS Capacitor: Threshold Voltage (2)

Potential balance: $V_{GB} = \phi_{ms} + \psi_s + \psi_{ox}$

$$V_T = \phi_{ms} + 2\phi_{Fp} + \frac{qN_a x_{d,max} - Q'_{ss}}{C'_{ox}}$$

Earlier, we had derived: $x_{d,max} = \sqrt{\frac{4\varepsilon_s \phi_F}{qN_a}}$

$$V_T = \phi_{ms} + 2\phi_{Fp} + \frac{qN_a\sqrt{\frac{4\varepsilon_s \phi_F}{qN_a}} - Q'_{ss}}{C'_{ox}}$$

For MOS with substrate p-type:

$$V_T = \phi_{ms} + 2\phi_{Fp} + \frac{\sqrt{4q\varepsilon_s N_a \phi_{Fp}} - Q'_{ss}}{C'_{ox}}$$

- Similarly, we can derive threshold voltage of MOS with substrate n-type:

$$V_T = \phi_{ms} - 2\phi_{Fn} - \frac{(\sqrt{4q\varepsilon_s N_d \phi_{Fn}} + Q'_{ss})}{C'_{ox}}$$

MOS Capacitor: Threshold Voltage (3)

For MOS with silicon p-type:

$$V_T = \phi_{ms} + 2\phi_{Fp} + \frac{\sqrt{4q\varepsilon_s N_a \phi_{Fp}} - Q'_{ss}}{C'_{ox}}$$

- For MOS with silicon n-type:

$$V_T = \phi_{ms} - 2\phi_{Fn} - \frac{(\sqrt{4q\varepsilon_s N_d \phi_{Fn}} + Q'_{ss})}{C'_{ox}}$$

Factors on which threshold voltage of MOS (long channel MOSFET) depends on:

1. Material of gate (will determine ϕ_{ms})
2. Material of semiconductor (will determine ϕ_{ms})
3. Doping of semiconductor
4. Material of oxide (ϵ_{ox})
5. Oxide Thickness
6. Parasitic Oxide Charges

MOS Capacitor: Threshold Voltage vs. Doping (1)

For MOS with silicon p-type:

$$V_T = \phi_{ms} + 2\phi_{Fp} + \frac{\sqrt{4q\varepsilon_s N_a \phi_{Fp}} - Q'_{ss}}{C'_{ox}}$$

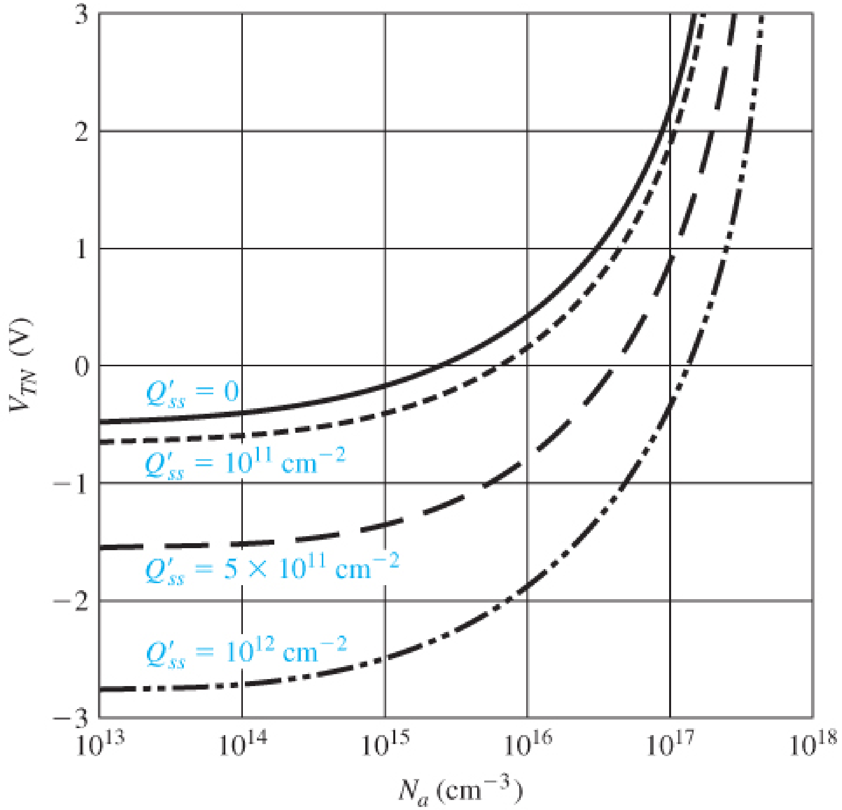


Figure 10.21 | Threshold voltage of an n-channel MOSFET versus the p-type substrate doping concentration for various values of oxide trapped charge (t_{ox} = 500 Å, aluminum gate).

MOS Capacitor: Threshold Voltage Vs. Doping (2)

- For MOS with silicon n-type:

$$V_T = \phi_{ms} - 2\phi_{Fn} - \frac{\left(\sqrt{4q\varepsilon_s N_d \phi_{Fn}} + Q'_{ss}\right)}{C'_{ox}}$$

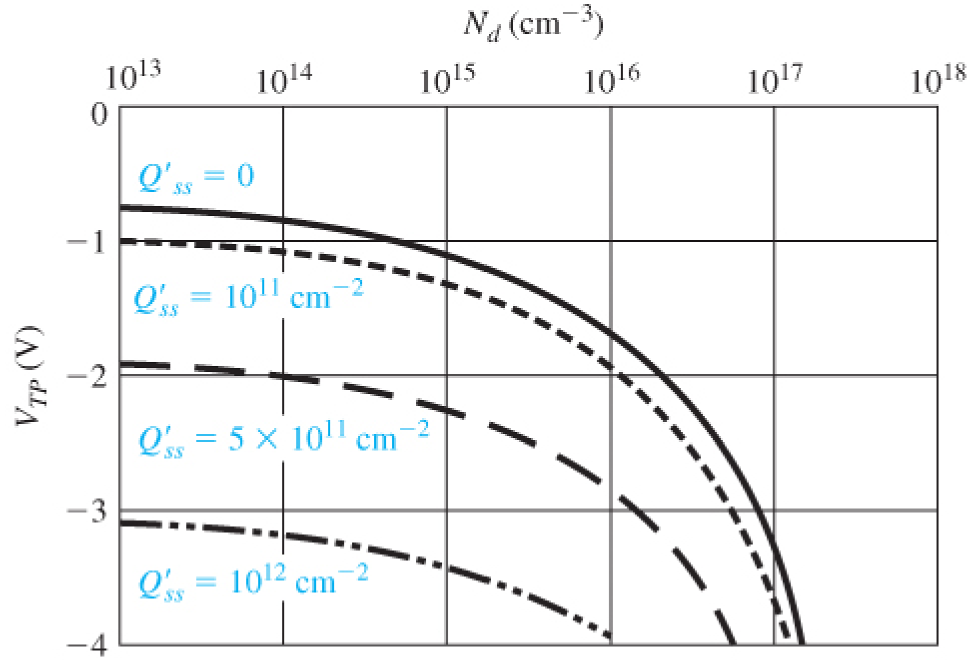


Figure 10.22 | Threshold voltage of a p-channel MOSFET versus the n-type substrate doping concentration for various values of oxide trapped charge (t_{ox} = 500 Å, aluminum gate).

References

- Tsividis, Yannis. *Operation and Modeling of the MOS Transistor.* McGraw-Hill, Inc., 1987.
- D. A. Neaman, *"Semiconductor Physics and Devices"*